Sector A branch-aware synthesis and termination package

TECT verification-first repository · claim A5-SECTOR-A-SYNTHESIS

first issued 2026-07-18 · this version issued 2026-07-18 · v1.0

1. Purpose and scope: referee question and review status

This package answers a narrow completion question: do the frozen Sector-A records form a reproducible path from definitions through full variational dynamics and spectral PDE approximation, together with controlled scalar perturbative and non-perturbative continuum limits? The answer is

$$\boxed{\texttt{PASS@BRANCH-AWARE-DECLARED-SCOPE.}} \tag{1.1}$$

The qualifier is load-bearing. The records form two compatible but distinct branches, not one parameter-identical full constructive theory. The first is the full-production variational/PDE/discretization branch. The second is the positive-mass real-scalar perturbative/constructive branch.

This v1.0 document is a T4 referee candidate. The primary and non-importing independent audits pass directly, but repository policy requires explicit operator review before this document may be confirmed, before the claim may be considered for scoped T5, and before a PUBLISHED bundle may be built. No pre-confirmation bundle exists.

2 Common definitions and conventions

The common spatial domain is the fixed spectral torus

$$\mathbb{T}^3 = [0, L_x) \times [0, L_y) \times [0, L_z), \qquad L_x = L_y = L_z = 16. \tag{2.1}$$

For a three-component complex field Ψ the real pairing is

$$\langle u, v\rangle_{\mathbb{R}} = \mathrm{Re} \int_{\mathbb{T}^3} u^\dagger v \, dx. \tag{2.2}$$

The full-production fourth-order scalar symbol inside each component is

$$K_{\mathrm{full}}(k) = r + Z|k|^2 + Y|k|^4 = m_{\mathrm{full}}^2 + Y(|k|^2 - q_0^2)^2, \tag{2.3}$$

where $Y = 1$, $Z < 0$, $q_0^2 = -Z/(2Y)$, and

$$m_{\mathrm{full}}^2 = r - \frac{Z^2}{4Y}. \tag{2.4}$$

The scalar-continuum branch uses the same Brazovskii form

$$K_{\mathrm{sc}}(k) = m_{\mathrm{sc}}^2 + Y(|k|^2 - q_0^2)^2 \tag{2.5}$$

with positive shell mass. Both branches use the spectral Fourier convention and $\eta_{\mathrm{shell}} = 0$ in the closed statements.

The local polynomial convention shared by the verified scalar reduction and the scalar Gibbs construction is

$$V(\phi) = \int_{\mathbb{T}^3} \left(\frac{\lambda}{4}\phi^4 + \frac{\gamma}{6}\phi^6 \right) dx, \qquad \gamma > 0. \tag{2.6}$$

The full P1 functional is stronger in field content, not in constructive scope: it has three complex components, family masses, an internal locking term, positive regularisers, and derivative Class-II currents in addition to the local polynomial.

3 The six frozen input results

The synthesis uses the following exact result boundaries.

D1: scalar kernel definition. `A1-PRODUCTION-KERNEL-MANIFEST` is T5. It pins the pure scalar Brazovskii symbol, the Fourier convention, q_0, Y, Z, shell mass, and $\eta_{\text{shell}} = 0$. It is a definition and implementation-identity result, not a phase-selection theorem.

D2: full variational realization. `A1-PRODUCTION-FUNCTIONAL-REALISATION` is T5 within its discrete matrix scope. For grids 4, 6, and 8, five field classes, and three finite- difference steps it verifies

$$DF[\Psi](v) = \langle R[\Psi], v \rangle_{\mathbb{R}}, \quad DR[\Psi]v = H[\Psi]v, \quad \langle u, Hv \rangle_{\mathbb{R}} = \langle Hu, v \rangle_{\mathbb{R}}. \tag{3.1}$$

It also verifies the local scalar reduction. This is the canonical standalone functional; it does not repair the historical non-variational solver source.

D3: full PDE theorem. `A2-FULL-PRODUCTION-WELLPOSED` is a T6 conditional theorem for the P1 functional on (2.1), with H^2 initial data, positive regularisers, and $\eta_{\text{shell}} = 0$. The linear operator is self-adjoint and bounded below; the coefficient conditions and positive sextic make the energy coercive; the gradient flow has a unique global H^2 solution, continuous finite-time dependence, exact energy dissipation, and smoothness at every positive time.

D4: spectral PDE approximation theorem. `A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM` is a T6 conditional theorem for the same P1/P2 branch. On the pinned initial H^2 balls and $0 < \tau \le t \le T$, it provides finite explicit solution-ball and residual constants and proves restarted exact-Galerkin convergence. It covers positive time and exact nonlinear projection. It does not provide useful historical N32/N64/N128 error bars.

D5: scalar perturbative cutoff removal. `A3-PERTURBATIVE-CONTINUUM-CORRELATORS` is a T6 conditional theorem. For the real scalar $d = 3$ positive-mass q^4 kernel, every connected perturbative amplitude converges at each fixed external momentum under the sharp spectral regulator. The statement is order-by-order, not resummed, and excludes genuine finite-difference Route B.

D6: scalar constructive cutoff removal. `A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE` is a T6 conditional theorem. For $m_{\text{sh}}^2 > 0$, $Y > 0$, real λ, and $\gamma > 0$ on (2.1), the covariance is trace class, all sharp spectral Galerkin Gibbs measures are normalizable, and the full sequence converges weakly on real L^2. Lifted densities converge in total variation and the declared smeared cylinder- polynomial correlations converge. This is finite volume and real scalar.

4 The branch graph and its proof

The exact dependency graph is

```
shared spectral geometry and named hypotheses
  |
  +-- full-production branch
  |      A1-PRODUCTION-FUNCTIONAL-REALISATION
  |        -> A2-FULL-PRODUCTION-WELLPOSED
  |        -> A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM
  |
  +-- scalar-continuum branch
         A1-PRODUCTION-KERNEL-MANIFEST
           +-> A3-PERTURBATIVE-CONTINUUM-CORRELATORS
           +-> A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE
```

The two scalar arrows are parallel. They record a shared positive-mass spectral anchor, not a claim that the A4 measure is derived from a finite perturbation series or that either result is a hard dependency of the other.

Full-production edge proof. The A2 card names the P1 functional as a hard dependency and carries `A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL`. The A3 full-discretization card names both P1 and P2 as hard dependencies and carries the same hypothesis. Its PDE residual, energy, solution ball, and exact-Galerkin flow therefore refer to the same canonical functional used in D3. This proves the two directed full-production edges without identifying a historical solver.

Scalar-continuum edge proof. D5 and D6 both state real-scalar, positive-mass, spectral/Galerkin scope. D5 uses the q^{-4} domination and fixed external momenta. D6 uses the same positive-mass q^4 covariance class plus the local quartic/sextic potential. The D6 theorem is general in the positive mass and audits the two production anchors as sanity checks. Thus D5 and D6 are compatible scalar continuum controls, while D6 remains logically non-perturbative.

5 The mass fork is necessary, not cosmetic

The canonical scalar kernel stores

$$m_{\mathrm{sc}}^2 = 0.005. \tag{5.1}$$

The full-production parameters store

$$r = 0.4740336473, \quad Z = -0.9252754126, \quad Y = 1,$$

so independent reconstruction gives

$$m_{\mathrm{full}}^2 = r - \frac{Z^2}{4Y} = 0.260000000009475. \tag{5.2}$$

Consequently

$$|m_{\mathrm{full}}^2 - m_{\mathrm{sc}}^2| = 0.255000000009475 > 0.2. \tag{5.3}$$

The separation is vastly larger than the upstream 10^{-9} kernel tolerance. It cannot be attributed to rounding. A4 applies at either positive mass, but that generality does not identify the two anchored models. Any synthesis that silently writes (5.1)=(5.2) is false.

6 The functional fork is also necessary

The P1 audit bounds the scalar-reduction relative error by

$$6.218564371786818 \times 10^{-16}. \tag{6.1}$$

This proves the local convention (2.6) inside the full functional when the non-scalar structures are switched off. It does not prove equality of the full and scalar measures. The full Class-II density contains derivative currents. A4 Gaussian samples have covariance regularity only in H^s for $s < 1/2$; the derivative-current squares are not automatically defined there. An extension needs a separate divergence and renormalisation analysis. No edge from D4 to a full Class-II constructive measure is present.

7 Synthesis proposition and proof

Proposition. Under the six registered named hypotheses and the exact scopes D1–D6, Sector A has: (i) a reproducible definitions-to-full-dynamics-to-spectral- approximation branch; (ii) a reproducible real-scalar spectral branch with both order-by-order and finite-volume non-perturbative cutoff removal; and (iii) an explicit interface contract that prevents parameter and functional identification across the two branches.

Proof. D1 and D2 fix the scalar and full functional definitions. The hard-dependency edges proved in Section 4 compose D2, D3, and D4 into the full-production branch. D5 and D6 provide the two distinct continuum statements on the scalar spectral class. Equations (2.1)–(2.6) verify the shared geometry and local convention. Equations (5.1)–(5.3) prove that the numeric models are not parameter-identical, while Section 6 proves that a scalar reduction is not a full-measure equivalence. The frozen manifest makes these edges and non-edges explicit. The two independent audits in Section 8 rehash the six records and reconstruct every load-bearing interface. This proves the proposition at the declared synthesis scope. ☐

8 Content addressing and direct reproduction

The synthesis manifest pins six status cards, four component manifests, two component evidence artifacts, and four existing PUBLISHED support-bundle manifests. The support-bundle content digests are

```
A1 scalar kernel:       99b255e767b5c1a6ea12983399f46ae5f23f5d3d76e4d587d3350c5c5ed62f7e
A1 full functional:    30749f1c5e1cdb9528ced2b6a3dcfced8d8ca8f6f39e8cef03913d32e13f33d6
A2 full PDE:           f07a39627a2eccc251fc67d1c988b9de18ec0b5643664fc60c3da0acc2eeeddb
A3 discretization:      6d15d165a73d3a2af07e10fce07394ce8b83311e571ba2aae2fbbc61c31d2e41
```

The primary audit checks all files listed in those support manifests, their recomputed digests, publication markers, and runlogs. The non-importing audit independently recomputes the manifest digests and reconstructs the dependency graph and mass fork using Decimal arithmetic.

Run from the repository root:

```
python codes/foundations/a5_sector_a_synthesis_verify.py
```

The required output is

```
PASS: primary (16/16)
PASS: independent (16/16)
ASSERTS: 32/32
A5-SECTOR-A-SYNTHESIS-INTEGRATED-PASS
Termination: PASS@BRANCH-AWARE-DECLARED-SCOPE
```

The wrapper re-executes both audits in a temporary directory, compares their branch and mass reconstructions, and writes only the integrated result to the declared evidence path. The direct primary and independent result artifacts are also retained for review.

9 Termination meaning

The result closes a documentation, interface, and reproducibility weakness in Sector A. In plain terms, the repository can now answer which mathematical model is defined, which full field equation is controlled, why its spectral approximation converges in the pinned positive-time scope, and what scalar continuum limits exist without perturbation theory. It also records exactly where those statements stop.

This is not the claim that energy underwent a phase transition and thereby created the present universe. No thermodynamic phase transition, vacuum selection, BCC existence, cosmology, or observation is proved here. The appropriate eventual result class is a scoped T5 synthesis closure, not a T6 new theorem and not T7 physical-domain closure.

10 Devil's-advocate review

1. **The requested P1–P4 sequence is one linear theory. UPHELD as false.** Sections 5 and 6 force a two-branch map.
2. **The two shell masses differ only through decimal rounding. DISMISSED.** Equation (5.3) is larger than 0.2, versus a 10^{-9} upstream tolerance.
3. **The A4 theorem constructs the full derivative Class-II measure. UPHELD as false.** D6 is a real-scalar local-polynomial construction, and the derivative currents require separate analysis.
4. **The P1 scalar reduction proves measure equivalence. UPHELD as false.** It proves a local functional identity on a slice, not equality after restoring internal fields and derivative currents.
5. **The component T6 theorems automatically make A5 a T6 theorem. UPHELD as false.** A5 is a meta-level synthesis and interface audit; it introduces no new mathematical theorem beyond D1–D6.
6. **Four published support bundles make A5 unnecessary. DISMISSED.** No individual bundle states or audits the cross-branch mass and functional non-implications.
7. **The graph secretly depends on the historical external solver. DISMISSED in the declared scope.** D2 uses the standalone canonical functional; the historical source remains an explicit excluded audit failure.
8. **P3 proves useful error bars for historical N32/N64/N128 runs. UPHELD as false.** D4 is an exact-Galerkin positive-time theorem with finite but extremely conservative constants and a distinct solver scope.
9. **Finite-volume cutoff removal proves a phase transition. UPHELD as false.** D6 contains no thermodynamic limit or spontaneous symmetry-breaking statement.
10. **This package closes BCC selection or Sector B. UPHELD as false.** BCC existence, selection, minimization, Sector B, and physical-domain closure are outside the proposition.
11. **Internal 32/32 execution permits immediate publication. UPHELD as false.** The binding workflow requires explicit operator review before confirmation, tier action, and final packaging.
12. **A synthesis package can omit a falsifier because it adds no physics. UPHELD as false.** Source drift, a broken dependency edge, or erased branch fork falsifies the declared synthesis result.

11 Falsifier and no-overclaim boundary

The synthesis is falsified by any failed source or evidence hash; a support bundle digest or runlog failure; a component record no longer ACTIVE and reproducible in its pinned scope; an unregistered hypothesis; a missing P1 to P2 to P3 hard-dependency edge; a failed 32-assertion verifier; equality or silent substitution of (5.1) and (5.2); or a valid proof that D6 already defines the full derivative Class-II measure contrary to its card.

The package does not cover the historical non-variational solver, a full three-component derivative Class-II constructive measure, $\eta_{\text{shell}} \neq 0$, removal of the ρ or Class-II mass regularisers, $t = 0$ algebraic P3 rates from merely H^2 data, practical historical-grid error bars, finite-difference Route B, infinite volume, phase transition, minimizer uniqueness, BCC existence or selection, Sector B, physical-domain closure, T6, or T7.

12 Review and packaging gate

The note, manifest, and direct scripts are the complete pre-confirmation review surface. The operator should attack the branch graph, equations (5.1)–(5.3), the functional non-implication, the six component scopes, and the 32/32 evidence. Corrections require a versioned re-issue. Only an explicit confirmation may close `A5-SECTOR-A-SYNTHESIS-OPERATOR-CONFIRMATION`. After that event, the confirmation marker and changelog record are added, scoped T5 is reviewed, and one claim-level PUBLISHED bundle is built last and integrity-checked.

Result footer

```
Result ID:               A5-SECTOR-A-SYNTHESIS
Precise statement:       Sector A has a reproducible full-production
                         variational/PDE/Galerkin branch and a separate
                         real-scalar perturbative/constructive branch,
                         with exact interfaces and non-implications frozen.
Scope:                   T4 REFEREE-CANDIDATE at branch-aware synthesis
                         scope; fixed spectral T3, eta_shell=0; mass and
                         functional forks retained.
Dependencies:            A1-PRODUCTION-KERNEL-MANIFEST;
                         A1-PRODUCTION-FUNCTIONAL-REALISATION;
                         A2-FULL-PRODUCTION-WELLPOSED;
                         A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM;
                         A3-PERTURBATIVE-CONTINUUM-CORRELATORS;
                         A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE;
                         six named hypotheses in the frozen manifest.
Evidence grade:          EXACT + EXECUTED + CONDITIONAL.
Reproduction command:    python codes/foundations/
                         a5_sector_a_synthesis_verify.py
Expected output:         primary 16/16; independent 16/16; ASSERTS 32/32;
                         A5-SECTOR-A-SYNTHESIS-INTEGRATED-PASS;
                         PASS@BRANCH-AWARE-DECLARED-SCOPE.
Falsification gate:      Hash/digest/runlog drift, broken dependency or
                         hypothesis, failed audit, or erased branch fork.
Tier before / after:     new / T4 REFEREE-CANDIDATE.
No-overclaim statement:  Not one parameter-identical full constructive
                         theory; not full Class-II measure, Route B,
                         infinite volume, phase transition, BCC, Sector B,
                         physical closure, T6, or T7.
Next required action:    Operator reviews and explicitly confirms or returns
                         corrections; only then build the PUBLISHED bundle.
```